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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : M-302

MATHEMATICS - III

Time Allotted : 3 Hours *Full Marks* : 70

The figures in the margin indicate full marks.

GROUP – A

(Short Answer Type Questions)

1. Answer any *ten* of the following : $10 \times 2 = 20$

i) If $f(x) = x^2$, $-\pi \le x \le \pi$ be represented in Fourier series as $\frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx)$ then find the value of b_n.

ii) A and B play a game in which their chances of winning are in the ratio 3 : 2. Find A's chances of winning three games out of the five games played.

iii) Find the Fourier sine transform of the function of $f(x) = 1$, $0 < x < \infty$.

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iv) Obtain the period of $\sin 3x$.

v) Find the bilinear transformation which maps the points $z = 1, 0, -1$ onto the points $w = i, 0, -i$.

vi) Expand $f(z) = \frac{1}{3z+2}$ in Taylor series about origin.

vii) Check the nature of the singularity of $f(z) = ze^{\frac{1}{z}}$ at $z = \infty$.

viii) If the random variable X takes the values 1,2,3 and 4 such that $P(X=1) = P(X=2) = 2P(X=3) = P(X+4)$, find the probability distribution of X.

ix) Find the residue of $f(z) = \frac{z}{(z-1)^4(z-2)(z-3)}$ at $z = 2$.

x) Let S be a sample space and two mutually exclusive events A and B be such the $A \cup B = S$. If $P(\ )$ denotes the probability of the event, then find the maximum value of $P(A) \times P(B)$.

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xi) A random variable follows Binomial distribution with mean 4 and standard deviation $\sqrt{2}$. Find the probability of assuming non-zero value of the variate.

xii) If $f(x) = kx$, $x = 1, 2, \ldots\ldots\ldots, n$ be the *p.m.f.* of a random variable x, then find the value of k.

xiii) A nationalized bank has found that the daily balance available in its savings a/c follows a normal distribution with mean Rs. 500 and s.d. Rs. 50. Find the percentage of savings a/c holders who maintain an average daily balance more than Rs. 500.

xiv) If two events A & B are independent, prove that $\overline{A}$ & B are also independent, where $\overline{A}$ is the complement of A.

xv) Assuming the recurrence relation for $J_n(x)$, prove that $J'_0(x) = -J_1(x)$.

xvi) Prove that $P_2(x) = \frac{1}{2}(3x^2 - 1)$.

GROUP – B

Answer any *five* questions taking at least *one* question from each Module. 5 × 10 = 50

Module – I

2. a) Find the Fourier cosine series for the function $f(x) = \pi - x, 0 \le x \le \pi$ and deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots = \frac{\pi^2}{8}$ 4 + 2

b) Find the inverse Fourier transform of the function $\frac{1}{s^2 + 6s + 25}$ 4

3. a) Find the Fourier transform of $f(x) = \begin{cases} 1, |x| < a \\ 0, |x| > a \end{cases}$ and hence find $\int_0^\infty \frac{\sin t}{t} dt$. 4 + 2

b) Find the Fourier sine transform of $e^{-|x|}$. 4

Module – II

4. a) The real part of an analytic function $f(z)$ where $z = x + iy$ is given by $e^{-y}\cos x$. Find the imaginary part of $f(z)$. 5

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b) Use Milne-Thomson method to find the analytic function $f(z) = u(x,y) + iv(x,y)$, where

$v(x,y) = x^4 - 6x^2y^2 + y^4$. 5

5. a) Use Cauchy's integral formula to evaluate $\int_C \frac{e^{2z}}{(z-1)(z-2)}dz$ where $C: |z| = 2$. 5

b) Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in Laurent's series valid for the region $|z| > 3$. 5

Module – III

6. a) In a bolt factory, machines *A*, *B*, *C* manufacture respectively 20%, 30% and 50% of the total output. Of the output, 3%, 4% and 5% are defective. One bolt is drawn at random from the product and found defective. What is the probability that it is manufactured by *A* ? 5

b) A continuous random variable *X* has the pdf $f(x) = \frac{1}{2} - ax, 0 \le x \le 4, a$ is a constant. Determine a and the probability that *X* lies between 2 and 3. 5

7. a) *P* is a point on the straight line *AB* of length $2a$. If *O* be the mid-point of *AB*, find the probability that *AO*, *AP*, *PB* form the sides of a triangle. 5

b) 50% of a product have weight 60 kg or more and 10% have weight less than 55 kg. Assuming the distribution of weight to be normal (μ, σ^2), evaluate $(\mu - 3\sigma, \mu + 3\sigma)$

[For a standard normal variate z, $P(z < 1.28) = 0.9$] 5

8. a) The incidence of occupational disease in an industry is such that workers have a 10% chance of suffering from it. What is the chance that out of 6 workers, at least 3 will suffer from the disease ? 5

b) If 3% of the bolts manufactured by a company are defective, find the probability that in a sample of 200 bolts (a) 5 will be defective, (b) none is defective. $[e^{-6} = 0.00248]$ 5

Module – IV

9. a) Using Fourier sine transform method, solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, (x > 0, t > 0)$ subject to the conditions $u(0,t) = 0, u(x,0) = \begin{cases} 1, 0 < x < 1 \\ 0, x \ge 1 \end{cases}$ & $u(x, t)$ is bounded. 7

b) Prove that $J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x$. 3

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10. a) Use the method of Frobenius to find the solution of the differential equation $9x(1-x)\frac{d^2y}{dx^2} - 12\frac{dy}{dx} + 4y = 0$ about $x = 0$. 7

b) Express $x^3 + x^2$ in terms of Legendre polynomials P_0, P_1, P_2, P_3. 3

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